

TREATMENT AND EMERGENCY RECOMMENDATIONS FOR Isobutyryl CoA Dehydrogenase Deficiency (IBD Deficiency)

PATIENT: _____ **MRN:** _____ **DOB:** _____

This patient has IBD Deficiency and is at risk for serious complications from this condition.

Definition: Isobutyryl Co-A dehydrogenase (IBD deficiency) is an inherited metabolic condition caused by the lack of enzyme isobutyryl Co-A dehydrogenase. This enzyme is part of the pathway utilized to break down protein, particularly valine. Individuals with IBD deficiency may have difficulty breaking down valine as well as utilizing dietary fat. This can result in production of abnormal organic acids and hypoglycemia (ketosis may be low or absent) during intercurrent illness. IBD deficiency is very rare and the clinical effects are not completely known. Some individuals may have cardiac involvement and low muscle tone. The effects of catabolism in conjunction with IBD deficiency are not well understood. While some individuals may have no problems when ill, avoidance of prolonged catabolism is advisable.

Outpatient Care: Mild illness should be evaluated by a physician and treated on an outpatient basis. Increase sugar containing clear liquid intake (fruit juice, Gatorade) and continue oral carnitine (if on carnitine supplementation). If unable to meet calorie needs orally, admit for IV fluids with glucose.

Emergency Room Protocol:

A. IV Treatment:

1. IV fluids with D10 to keep blood glucose elevated. Do NOT wait for results of laboratory evaluation before starting IV fluids with glucose.
 - i. Start D10 with age appropriate electrolytes at 1.5 x maintenance immediately.
 - ii. If D10 is not available, use D5 until D10 can be obtained. Do not wait for D10 before starting IV fluids.
 - iii. Use D25, 2 ml/kg for hypoglycemia.
2. Carnitine supplementation – if patient does not tolerate oral carnitine, use IV carnitine at the home dose and schedule.

B. Laboratory studies:

1. Monitor blood glucose levels and other parameters as clinically indicated.
2. Electrolytes (Na, K, Cl, HCO₃)
3. Liver function tests
4. CBC+diff (add blood cultures if febrile)
5. Urinalysis
6. Plasma total and free carnitine levels
7. Plasma acylcarnitine profile
8. Urine organic acids
9. If altered mental status, serum ammonia level (draw from free flowing vein and place directly on ice. RUN STAT. Do not send capillary sample)

C. Evaluation and Follow-up:

1. Treat the underlying and precipitating illness (infection, etc), and obtain appropriate additional laboratory studies to establish the cause of the physiological challenge.
2. **Notify the metabolic specialist immediately (after business hours this number allows you to speak with the geneticist on call:**

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